Chuyu Xu

cxu433@gatech.edu · github.com/chuyuxu · Homepage

Research interests: LLM agents, with a focus on evaluation, forecasting, and automatic agent evolution

Education

Georgia Institute of Technology Aug 2025 – Dec 2026 (expected)
M.S. in Computational Science and Engineering GPA: 3.75/4.00 *Atlanta, GA*
Selected coursework: High Performance Computing, Computational Data Analysis, Graphical Models in ML, Computer Vision

Wuhan University Sep 2020 – Jun 2024
B.S. in Financial Engineering GPA: 3.83/4.00 *Wuhan, China*
Thesis: *Tree-Based Sorts: Predicting Future Returns Using Past Returns of A-shares*, advised by Prof. Bin Li

Publications & Preprints

[1] **EPSBench: Benchmarking LLM Agents for Earnings Forecasting**
Yanbin Yin*, **Chuyu Xu***, Liqin Ye, Jing Peng, Chao Zhang. *Under review at ICLR 2027.* [Code] [arXiv:2610.38293]

[2] **GenHarness: General and Automatic Harness Evolution Across Domains**
Yanbin Yin, Xiang Chen, Tong Yu, Mehrab Tanjim, **Chuyu Xu**, Chao Zhang, Saayan Mitra. *Under review at NAACL 2027*

[3] **CoLoR: Compact Low-Rank Residuals for Host-Agnostic Diffusion Transformer Caching**
Yifan Jia, Zhiyuan Chen, Yuxiang Zhao, **Chuyu Xu**, Yufei Ma. *Under review at AAAI 2027*

*Equal contribution.

Research Experience

Graduate Researcher, advised by Prof. Chao Zhang Dec 2025 – Present
School of Computational Science and Engineering *Georgia Tech*

Agent forecasting — EPSBench [1]

- Led the design of a benchmark (149 firm-quarters, 86 US companies) in which LLM agents estimate income-statement drivers with cited evidence and a shared calculator derives EPS, making each forecast assumption checkable and each error traceable to a line item.
- Built the point-in-time environment: an LLM pipeline compiles SEC filings into per-firm accounting schemas linking drivers to EPS; MCP tools over filings, transcripts, and news are cut off the day before each release; and a memorization test finds no detectable recall.
- Designed the evaluation framework (price-deflated error, paired bootstrap tests against IBES analyst consensus, proper scoring rules for intervals) and ran 6,000+ agent runs across 6 models plus harness and information ablations.
- Found that leading agents show no detectable difference from analyst consensus in point error, yet their 80% intervals cover only 37–70% of outcomes, with most misses above the upper bound as forecasts run low; model gaps concentrate in operating income.

Agent harness evolution — GenHarness [2], in collaboration with Adobe Research

- Contributed to developing a framework that compiles lightweight domain configurations into harness-evolution workflows, raising average performance from 42.9% to 52.6% over strong vanilla harnesses across four domains.
- Designed the system overview figure, wrote the related-work section, and refined the full manuscript.

Industry Experience

Peak Performance Technology, *Machine Learning Engineer Intern, Shanghai* Mar 2025 – Jul 2025

- Raised minority-class recall above 95% for a ResNet-ViT acoustic classifier via data-driven resampling.

Apple, *Data Analyst Intern, Shanghai* Jun 2024 – Nov 2024

- Built a gradient-boosting model for conversion-rate forecasting (R^2 = 0.81), outperforming the incumbent forecast by 50,000 units.
- Built a fraud-detection system (rules + Isolation Forest/autoencoder), cutting unauthorized distribution by 70%.

Sea Level Capital, *Quantitative Researcher Intern, Shanghai* Jan 2024 – May 2024

- Developed Random Forest and XGBoost models for equity return forecasting from time-series features.

Jialiang Capital, *Quantitative Researcher Intern, Wuhan* Jun 2023 – Oct 2023

- Designed 8 statistically motivated predictive features for equity return forecasting.

Technical Skills

LLM & Agents: agent harnesses (Claude Code, OpenCode), MCP tool integration, LLM-based data pipelines
Evaluation: benchmark design, contamination analysis, statistical significance testing, uncertainty calibration
Programming & Compute: Python, SQL, PyTorch, Git, Slurm (HPC GPU clusters)